ZS-S10

Gauge Bridge via Stückelberg-Corollary IV Mechanism

Action-Level Closure of the U(1)_Z ↔ U(1)_Y Identification and the Vortex-Core 4π Spinor Realization

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§0. Abstract

Z-Spin Cosmology v1.0 establishes the Standard Model gauge group structure (ZS-M9), gauge coupling constants (ZS-S1), fermion mass ratios (ZS-M10, ZS-M11), the Higgs VEV (ZS-S4 §6.12), and the electric charge spectrum (ZS-U9). The compact U(1)_Z gauge symmetry of the Z-bias field Φ (ZS-F1 §3.2, PROVEN) and the Standard Model U(1)_Y hypercharge gauge group have both been present in the corpus, but their action-level identification was flagged as Gap G1 of the Trinity Braiding Theorem (ZS-U9 NC-U9.2). Separately, the Horizon Spinor Theorem (ZS-A7 §3, DERIVED via Theorem 3.2-bis) established that the boundary holonomy operator B_Z at any Z-Spin event horizon carries a strict 4π closure period, and Corollary IV (ZS-A7 §4.4, DERIVED) identified the Bose/Fermi duality of Z-anchored vortices — but an action-level realization was absent.

This paper introduces the **ZS-S10 master action**, a minimal extension of the ZS-F1 action obtained by promoting the partial derivative of Φ to a gauge-covariant derivative $D_\mu\Phi = (\partial_\mu - i\kappa g_Y B_\mu)\Phi$ coupled to the Standard Model hypercharge gauge field B_μ, with mixing scale $\kappa^2 = A/Q = 35/4807$ derived from the Register-Total Normalization Theorem (ZS-M6 Theorem 2.2.1, DERIVED). **No new parameters, no new fields, no new postulates** are introduced beyond this minimal coupling. The master action simultaneously closes Gap G1 (via definitional Stückelberg-type identification at the action level) and provides the action-level field-theoretic realization of the vortex Bose/Fermi duality (Corollary IV).

The paper establishes five principal results. **Theorem S10.1 (Stückelberg Mixing Scale, DERIVED-CONDITIONAL)** fixes the mixing scale $f^2 = \kappa^2 M^2_P = (A/Q)M^2_P$ at numerical uniqueness precision 10^{-14}%. **Theorem S10.2 (L_XY=0 Preservation, PROVEN-PERTURBATIVE)** extends the ZS-M6 §7A Continuum Perturbative Protection Theorem to the ZS-S10 action through a five-step proof. **Theorem S10.3 (Stückelberg-Corollary IV Bridge, DERIVED)** establishes that the UV limit (vortex core, $r \sim \ell_P$) realizes the (F) branch of Corollary IV with the j=1/2 spinor structure, while the IR limit (galactic scale, $r \gg \lambda_C(B)$) realizes the (B) branch with 2π Goldstone winding. **Theorem S10.4 (Φ Field U(1)_Y Charge, DERIVED)** fixes $q_\Phi = +1$ uniquely through four independent constraints, three of which are independent of ZS-S10 itself, removing the apparent circularity of the Gap G1 self-closure. **Theorem S10.5-BPS (Bogomolnyi BPS Spinor Lift, DERIVED)** provides a seven-step explicit proof that the ZS-A7 Theorem 3.2-bis Kraus-operator 4π closure emerges naturally from the action content of ZS-S10 at the Z-anchored vortex core, with observable realization $\tilde{u}_{seam}(\alpha) = \cos(\alpha/2)$.

The master action preserves all backward compatibility with the existing corpus at action level: $B_\mu = 0$ recovers ZS-F1 exactly; galactic scales ($r/\lambda_C(B) \sim 10^{53}$) recover ZS-A1 78/78 PASS through exponential Stückelberg screening; inflation ($|\Phi| \gg 1$) recovers ZS-U1 r = 0.0089 unchanged; vortex core realizes

ZS-A7 Cor IV (F) with ZS-A6 §4.5.6 cigar bounce profile; galactic halo realizes ZS-A7 Cor IV (B) with ZS-A1 §2 Goldstone isothermal profile. Gap G1 (Trinity Braiding Theorem, ZS-U9 §8) is upgraded from OPEN to **CLOSED** at the action level; Gap G2 (ZS-M9 Table 2 assignment upgrade) is advanced via the anomaly-preserving structure of Theorem S10.2.

Keywords: Stückelberg mechanism, gauge bridge, U(1)_Z, U(1)_Y, hypercharge, Z-bias field, vortex Bose/Fermi duality, 4π closure, Bogomolnyi vortex, Kraus operator, signed seam witness, Z-Spin Cosmology, zero free parameters

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§0.1 Epistemic Status Legend

Status	Definition
LOCKED	Core constant derived and fixed in an upstream paper; no downstream paper may modify.
PROVEN	Mathematical theorem; follows from standard mathematics or from corpus definitions alone. Machine-verifiable.
DERIVED	Follows from the Z-Spin action plus PROVEN inputs. Zero free parameters beyond A = 35/437.
DERIVED-CONDITIONAL	Derived from Z-Spin axioms, conditional on a stated upstream assumption.
PROVEN-PERTURBATIVE	Theorem valid to all orders in Lorentz-invariant perturbation theory; non-perturbative/strong-curvature regime not covered.
VERIFIED	Numerically confirmed against observational data or independent computation.
TESTABLE	Well-defined prediction awaiting experimental data.
HYPOTHESIS	Physically motivated conjecture; derivation chain incomplete.
OBSERVATION	Numerical proximity confirmed with anti-numerology tests; no action-level derivation yet.
CONSISTENT	Compatible with framework structure but not independently derived.
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim.
RETRACTED	Previously claimed, now withdrawn with documented reason.

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§1. Introduction

1.1 Motivation: Gap G1 of the Trinity Braiding Theorem

The Trinity Braiding Theorem (ZS-U9 v1.0 §6, DERIVED conditional on gaps G1, G2) derives the complete Standard Model hypercharge spectrum $Y_Q = +1/6$, $Y_u = +2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$, $Y_{\nu R} = 0$ from four ingredients: Compact Phase Integer Lattice (ZS-F1 §3.2, PROVEN), Yukawa Gauge-Lift (ZS-M10 Theorem 2.1, PROVEN), McKay SU(5) Cartan (ZS-M9 §5.2, DERIVED), and Neutral-Higgs Hypercharge Fixing (Theorem T3, DERIVED via ZS-S4 §6.12 Higgs VEV and photon masslessness). All five Standard Model anomaly cancellation conditions PASS automatically at integer precision (ZS-U9 §7 A1-A5).

Of the three originally open gaps, Gap G3 (compact phase normalization matching 1/6 Cartan factor) was closed by Theorem T3 in the dated update of April 19, 2026. Gap G2 (ZS-M9 Table 2 assignment upgrade from HYPOTHESIS strong to PROVEN) remains in progress. **Gap G1** — the action-level identification of $U(1)_Z$ (the Z-bias field gauge symmetry of ZS-F1 §3.2, PROVEN) with the Standard Model $U(1)_Y$ hypercharge gauge group — has been outstanding since the original v1.0 release.
The present paper closes Gap G1. The closure is achieved through a minimal extension of the ZS-F1 action, obtained by promoting the partial derivative $\partial_\mu\Phi$ to a gauge-covariant derivative $D_\mu\Phi = (\partial_\mu - i\kappa g_Y B_\mu)\Phi$, where B_μ is the Standard Model hypercharge gauge field, g_Y is its coupling, and $\kappa = \sqrt{(A/Q)}$ is the cross-coupling strength already established by the Register-Total Normalization Theorem (ZS-M6 Theorem 2.2.1, DERIVED; 10-step derivation chain C1-C10). No new parameters, no new fields, no new postulates are introduced.

1.2 Second Motivation: Action-Level Realization of Corollary IV

A separate corpus development provides the second motivation for ZS-S10. The Horizon Spinor Theorem (ZS-A7 §3, DERIVED via Theorem 3.2-bis, PROVEN) established that the boundary holonomy operator B_Z at any Z-Spin event horizon carries a strict 4π closure period, inherited from the $j = 1/2$ representation theory of the Z-sector mediator (ZS-M3 Theorem 5.1 uniqueness). Corollary IV (ZS-A7 §4.4, DERIVED) then identified the Bose/Fermi duality of any Z-anchored vortex: the core (Region I) hosts the 4π spinor closure, while the exterior (Region II) hosts the 2π Goldstone winding of $\pi_1(U(1)) = \mathbb{Z}$.
While the topological structure of Corollary IV is fully established at the corpus level, its **action-level realization** — an explicit demonstration that both the (F) and (B) branches emerge from a single field-theoretic action as two limits of the same field content — was absent. The ZS-S10 master action provides this realization: the vortex core ($|\Phi| \to 0$, $\rho \to 0$ limit) realizes the (F) branch, while the galactic scale ($|\Phi| \approx 1$ with massless Goldstone θ) realizes the (B) branch. Both limits share the common Z-Anchor boundary condition $|\Phi(x_0)| = 0$ (ZS-F1 §5.2 PROVEN, ZS-A6 §4.5.6 cigar bounce realization DERIVED).

1.3 Scope and Structure

The paper is organized as follows. §2 collects all locked inputs from the prior corpus, including $A = 35/437$, $Q = 11$, $\kappa^2 = A/Q$, the ZS-F1 action, the ZS-A7 Theorem 3.2-bis Kraus structure, and the ZS-A6 cigar vortex profile. §3 presents the ZS-S10 master action and establishes Theorem S10.1 (Stückelberg mixing scale). §4 proves Theorem S10.2 ($L_{XY} = 0$ preservation). §5 establishes Theorem S10.3 (Stückelberg-Cor IV Bridge) with explicit backward-compatibility verification in three regimes ($B_\mu = 0$, galactic, inflation). §6 presents Theorem S10.4 (Φ Field $U(1)_Y$ Charge) with the four-path circularity resolution. §7 presents Theorem S10.5-BPS (Bogomolnyi BPS Spinor Lift) with the seven-step explicit derivation. §8 registers the non-claims and the gate updates. §9 summarizes the derivation chain and verification matrix. Appendices provide the verification suite results (Appendix A), cross-reference table (Appendix B), and the companion Python verification script (Appendix C).

1.4 Principal Contribution

The principal contribution of ZS-S10 is the identification that **a single minimal coupling modification** of the ZS-F1 action — promoting $\partial_\mu\Phi$ to $D_\mu\Phi = (\partial_\mu - i\kappa g_Y B_\mu)\Phi$ — simultaneously achieves:
(a) Closure of Gap G1 at the action level (Theorem S10.1 + Theorem S10.4);
(b) Action-level realization of Corollary IV (F) and (B) branches (Theorem S10.3);

(c) Explicit Kraus-operator 4π closure at the vortex core (Theorem S10.5-BPS);
(d) Preservation of L_XY = 0 sector independence to all perturbative orders (Theorem S10.2);
(e) Full backward compatibility with ZS-F1, ZS-A1, ZS-U1, ZS-A7 at the action level.
The five results converge on a single structural statement: the U(1)_Z of ZS-F1 and the U(1)_Y of the Standard Model are **the same gauge group**, coupled to the Z-bias field Φ with charge q_Φ = +1 (Theorem S10.4 DERIVED) and to Standard Model fermions with the ZS-U9 Trinity Braiding hypercharges (Y_Q, Y_u, Y_d, Y_L, Y_e, Y_νR), through a minimal coupling whose scale is fixed by the per-mode cross-coupling strength $\kappa^2 = A/Q$ already established in ZS-M6. No new physics, no tuning, no hierarchy problems — only the integration of two existing corpus structures into a single action-level object.

§2. Locked Inputs

All inputs to ZS-S10 are LOCKED, PROVEN, or DERIVED in the prior corpus. The table below isolates the inputs used in this paper; a complete cross-reference is given in Appendix B.

2.1 Foundational Constants

Table 2.1. Locked Z-Spin constants.

Quantity	Value	Source	Status
A (geometric impedance)	35/437 = 0.080092	ZS-F2 v1.0 §7	**LOCKED**
(Z, X, Y) sector dimensions	(2, 3, 6); Q = Z+X+Y = 11	ZS-F5 v1.0 §4	**PROVEN**
G = MUB(Q)	Q+1 = 12	ZS-F5 v1.0	**PROVEN**
κ^2 (per-mode cross-coupling)	A/Q = 35/4807 ≈ 0.007281	ZS-M6 Theorem 2.2.1	**DERIVED**
δ_X (X-sector asymmetry)	5/19	ZS-F2 v1.0	**PROVEN**
δ_Y (Y-sector asymmetry)	7/23	ZS-F2 v1.0	**PROVEN**
λ_vac (vacuum self-coupling)	$2A^2 = 0.01283$	ZS-U5 v1.0 §15.4	**DERIVED-CONDITIONAL**
m_ρ (radial mode mass)	2A·M_P = 0.1602 M_P	ZS-F1 v1.0 §4.4	**DERIVED**
M_P (reduced Planck mass)	2.435×10^{18} GeV	Standard	**STANDARD**

2.2 ZS-F1 Base Action and U(1)_Z Structure

The Z-Spin base action in the Jordan frame [ZS-F1 v1.0 §3.1 PROVEN]:

$$S_{F1} = \int d^4x\sqrt{-g}\left[\frac{1}{2}M_P^2\left(1 + A|\Phi|^2\right)R - \frac{1}{2}M_P^2|\partial\Phi|^2 - V(\Phi)\right] + S_m$$

with $V(\Phi) = (\lambda/4)M^4_P(|\Phi|^2 - 1)^2$ and $\Phi(x) = \rho(x)\cdot\exp(i\theta(x)) \in \mathbb{C}$.
The U(1)_Z gauge symmetry [ZS-F1 v1.0 §3.2 PROVEN, direct quotation]: "The action is invariant under $\Phi \to \exp(i\alpha)\Phi$ for constant $\alpha \in [0, 2\pi)$." The compact domain $\alpha \in [0, 2\pi)$ is structurally essential (ZS-U9 §3.1): the Z-bias field Φ takes values in $\mathbb{C}$, the natural U(1) action is rotation by α modulo 2π, and the vacuum manifold $M = \{|\Phi| = 1\} \cong S^1$ has $\pi_1(S^1) = \mathbb{Z}$ (ZS-F1 §5.1, PROVEN).

2.3 Register-Total Normalization Theorem

From ZS-M6 Theorem 2.2.1 [DERIVED, 10-step chain C1-C10]:

$$\kappa^2 = \frac{A}{Q} = \frac{35}{4807} \; [EXACT\ RATIONAL]$$

Numerical uniqueness test (ZS-M6 §2.2.2): Among candidates A/Q, A/(Q−Z) = 35/3933, 3A/Q² = 105/52877, and A alone, κ² = A/Q is uniquely selected at 10^{-14}% precision, with alternative candidates deviating by 1.03×10^{-2}, 3.48×10^{-2}, and 3.67×10^{-1} respectively (226×, 765×, 8074× worse than A/Q).

2.4 L_XY = 0 Sector Independence

The block-Laplacian X-Y block vanishes identically:

$$L_{XY} \equiv 0 \quad [PROVEN,\ ZS-F1\ v1.0\ §9,\ ZS-S1\ v1.0\ §4,\ ZS-M6\ §2.2]$$

This is enforced at three independent levels: (i) the algebraic identity [su(2)_X, su(2)_Y] = 0 from the Lorentz algebra decomposition so(1,3) ⊗ ℂ ≅ su(2)_A ⊕ su(2)_B (ZS-M2 §2, PROVEN); (ii) the action-level absence of direct X-Y coupling terms in (1+A|Φ|²)R (ZS-F1 §9, PROVEN); (iii) the A_5-equivariant Schur protection via the Adjoint Obstruction Theorem (ZS-F2 §4.2A, PROVEN). The Continuum Perturbative Protection Theorem (ZS-M6 §7A, PROVEN-PERTURBATIVE) extends L_XY = 0 to all orders in perturbation theory.

2.5 ZS-A7 Kraus Operator Structure

The Z-mediated CPTP channel [ZS-Q1 §3.3, PROVEN] has exactly two Kraus operators:

$$\Lambda(\rho_X) = \sum_{z\in\{0,1\}} K_z \rho_X K_z^\dagger, \quad K_0 = P_{|4\rangle}, \quad K_1 = P_{|6\rangle}$$

where the Kraus index z runs over the Z-sector basis {|4⟩, |6⟩} of the Q = 11 register (ZS-A4 Appendix A). By ZS-A7 Theorem 3.2-bis [PROVEN]:

Part 1 (Single-factor structure): $K_z(\theta) = U_Z(\theta)_{zz'} K_{z'}(0)$ with $U_Z(\theta) = exp\left(-\, i\theta\sigma_y/2\right)$.

Part 2 (Choi-state period): $C_\Lambda(\theta + 2\pi) = C_\Lambda(\theta)$ (bilinear, 2π-periodic).

Part 3 (Signed seam witness period): $\tilde{u}_{seam}(\theta + 2\pi) = -\,\tilde{u}_{seam}(\theta)$, $\tilde{u}_{seam}(\theta + 4\pi) = +\,\tilde{u}_{seam}(\theta)$ (linear in Kraus, 4π-periodic).

Part 4 (Uniqueness): The signed seam witness is the unique linear-in-Kraus observable carrying the 4π signature.

2.6 ZS-A6 Cigar Vortex Profile

The Jaffe-Taubes anchored vortex [ZS-A6 §4.5.3, PROVEN; §4.5.6 DERIVED, 140/140 PASS] on the Euclidean Schwarzschild near-horizon geometry satisfies the cigar vortex equation:

$$f''(\rho) + \frac{1}{\rho} f'(\rho) - \frac{f}{\rho^2} - \lambda^* f\left(f^2 - 1\right) = 0$$

with $f(0) = 0$ (Z-Anchor) and $f(\rho \to \infty) \to 1$ (vacuum attractor). The Frobenius slope at the tip: $c_{cigar} = f'(0) = 0.06605585$ [ZS-A6 §4.5.6.2]. Wick-rotation match with the independent D1 EF-BVP: $c_{EF} = 2\sqrt{r_H} \cdot c_{cigar} = 0.93417$ vs $c_1 = 0.93500$ (0.089% agreement, ZS-A6 §4.5.6.4-5). **Existence and uniqueness** in the n = 1 sector is guaranteed by Theorem C3 (Jaffe-Taubes 1980, cited ZS-A6 §4.5.3).

2.7 ZS-U9 Hypercharge Spectrum

From ZS-U9 Theorem 6.1 [DERIVED, conditional on G1, G2]:

Table 2.2. Standard Model hypercharge spectrum.

Field	Y	Source
Q_L (quark doublet)	+1/6	ZS-U9 Theorem 6.1
u_R	+2/3	ZS-U9 Theorem 6.1
d_R	−1/3	ZS-U9 Theorem 6.1
L_L (lepton doublet)	−1/2	ZS-U9 Theorem 6.1
e_R	−1	ZS-U9 Theorem 6.1
ν_R	0	ZS-U9 Theorem 6.1
H (Higgs)	+1/2	ZS-U9 Theorem T3

All five Standard Model anomaly cancellation conditions A1-A5 PASS automatically at integer precision (ZS-U9 §7).

2.8 ZS-M3 j = 1/2 Uniqueness and SU(2) Sign Flip

Theorem 5.1 (ZS-M3, PROVEN): Among all half-integer spins j, dim(Inv_4(j)) = 2 = Z if and only if j = 1/2.

Lemma 10.1 (ZS-M3, PROVEN): $D^j(-I) = (-1)^{2j} \cdot I$. For j = 1/2: $D^{1/2}(-I) = -I$ (SU(2) double-cover sign flip).

2.9 Standard Model Gauge Couplings

From ZS-S1 v1.0 [PROVEN/DERIVED]:

$$\alpha_s = \frac{Q}{(V+F)_Y + \beta_0(Z)} = \frac{11}{93} = 0.11828\ (+\ 0.31\sigma\ vs\ PDG\ 2024)$$

$$sin^2\theta_W = \frac{48}{91} \cdot x^* = 0.23118\ (-\ 1.26\sigma\ vs\ PDG\ 2024)$$

where $x^* = Re(z^*) = 0.4383$ (ZS-M1 v1.0). The hypercharge gauge coupling at low energy satisfies $g_Y = e/cos\theta_W$ with $e^2 = 4\pi\alpha_{EM}$; combined with ZS-M8 NLO c_4 = 4/13 yielding $1/\alpha_{EM} = 137.0359$ (1.07 ppm from CODATA 2022), this gives $g_Y \approx 0.345$ at low energy (running to matching scale is discussed in §3.4).

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§3. The ZS-S10 Master Action and Theorem S10.1

3.1 Master Action

Definition 3.1 (ZS-S10 Master Action). The Z-Spin action of ZS-F1, extended by promotion of the partial derivative to a gauge-covariant derivative coupling the Z-bias field Φ to the Standard Model hypercharge gauge field B_μ, is:

$$S_{S10} = \int d^4x\sqrt{-g}\left[\frac{1}{2}M_P^2\left(1 + A|\Phi|^2\right)R - \frac{1}{2}M_P^2\left|D_\mu\Phi\right|^2 - V(\Phi) - \frac{1}{4}B_{\mu\nu}B^{\mu\nu}\right] + S_m[g, \psi, B]$$

where:

$$D_\mu\Phi = \left(\partial_\mu - i\kappa g_Y B_\mu\right)\Phi, \quad \kappa = \sqrt{A/Q} = \sqrt{35/4807}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu$$

$$V(\Phi) = (\lambda/4)M_P^4\left(|\Phi|^2 - 1\right)^2 \quad [unchanged\ from\ ZS - F1]$$

and $S_m[g, \psi, B]$ is the standard Standard Model matter action with fermion hypercharge couplings $\bar{\psi}_f i\gamma^\mu\left(\partial_\mu - ig_Y Y_f B_\mu\right)\psi_f$ using the ZS-U9 Trinity Braiding hypercharges Y_f of Table 2.2.

Minimal Extension Principle. The master action differs from the ZS-F1 action only in the replacement $\left|\partial_\mu \Phi\right|^2 \to \left|D_\mu \Phi\right|^2$ and the addition of the standard U(1)_Y gauge kinetic term $-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}$ (which already appears in S_m implicitly as the U(1)_Y sector of the Standard Model). No new fields beyond those already in ZS-F1 + Standard Model are introduced. No new parameters beyond A = 35/437 (LOCKED) are introduced: κ is determined from A and Q via ZS-M6 Theorem 2.2.1 (DERIVED), and g_Y is the standard SM hypercharge coupling.

3.2 Theorem S10.1: Stückelberg Mixing Scale

Theorem S10.1 (Stückelberg Mixing Scale, DERIVED-CONDITIONAL). The effective Stückelberg mixing scale f in the ZS-S10 master action satisfies:

$$f^2 = \kappa^2 M_P^2 = \frac{A}{Q} M_P^2 = \frac{35}{4807} M_P^2$$

Numerically: $f = 0.08533 \cdot M_P = 2.078 \times 10^{17}$ GeV.

Proof.

Step 1 (Expansion of |D_μΦ|²). Writing Φ = ρ·exp(iθ), the gauge-covariant kinetic term expands as:

$$\left|D_\mu \Phi\right|^2 = \left(\partial_\mu \rho\right)^2 + \rho^2\left(\partial_\mu \theta - \kappa g_Y B_\mu\right)\left(\partial^\mu \theta - \kappa g_Y B^\mu\right)$$

At the vacuum attractor ρ = 1 [ZS-F1 §4.2 DERIVED]:

$$\left|D_\mu \Phi\right|^2\Big|_{\rho=1} = \left(\partial_\mu \theta - \kappa g_Y B_\mu\right)\left(\partial^\mu \theta - \kappa g_Y B^\mu\right)$$

Multiplying by the kinetic prefactor $-\frac{1}{2}M_P^2$:

$$L_{kin,\,gauge}\Big|_{\rho=1} = -\frac{1}{2}M_P^2\left(\partial_\mu \theta - \kappa g_Y B_\mu\right)^2$$
$$= -\frac{1}{2}M_P^2(\partial\theta)^2 + M_P^2 \kappa g_Y B_\mu \partial^\mu \theta - \frac{1}{2}M_P^2 \kappa^2 g_Y^2 B_\mu B^\mu$$

Step 2 (Identification of Stückelberg mass). The last term provides the Stückelberg mass for B_μ:

$$m_B^2 = M_P^2 \kappa^2 g_Y^2 = \frac{A}{Q} g_Y^2 M_P^2$$

Reading this against the canonical Stückelberg Lagrangian $-\frac{1}{2}f^2\left(\partial_\mu \theta - g_Y B_\mu\right)^2$ at effective charge q = 1 identifies:

$$f^2 = \kappa^2 M_P^2 = \frac{A}{Q} M_P^2$$

Step 3 (Register-Total Normalization theorem basis). The mixing scale f² = (A/Q)M²_P is determined by the per-mode cross-coupling strength κ² = A/Q established in ZS-M6 Theorem 2.2.1 via the 10-step derivation chain C1-C10 (ZS-F1 non-minimal coupling → Regge discretization → Mode-Count Collapse → Gilkey heat kernel factorization → Spectral asymmetry → Product structure → Register-Total Normalization → Peter-Weyl/Schur orthogonality → rank-1 β_0-selected structure → Dimensional Coupling Norm). Each step is independently PROVEN, STANDARD, or LOCKED.

Numerical uniqueness (inherited from ZS-M6 §2.2.2): Among the natural candidates for κ², A/Q = 35/4807 is uniquely selected at 10^{-14}% precision. Alternative candidates (A/(Q−Z), 3A/Q², A alone) deviate by 226×, 765×, and 8074× respectively.

[STATUS: DERIVED-CONDITIONAL] Conditional on ZS-M6 Theorem 2.2.1 caveats R-1, R-2, R-3 (absolute 1-loop normalization from continuum action; register-scalar assumption; rank-1 from action level). These caveats do not affect the numerical content and are fully inherited from ZS-M6.

3.3 Physical Interpretation

The mixing scale f ≈ 2.08 × 10^{17} GeV sits at approximately 0.085 M_P, well above any Standard Model energy scale and below the Planck scale. This places f near the GUT unification scale. Combined with the lepton hypercharge Y_L = −1/2 (ZS-U9 Theorem 6.1), the Stückelberg mass of the B_μ gauge boson as felt by the lepton doublet is:

$$m_B^{(L)} = g_Y \cdot \left|Y_L\right| \cdot f \approx 0.345 \times 0.5 \times 2.08 \times 10^{17}\, GeV \approx 3.58 \times 10^{16}\, GeV$$

This is safely above all direct laboratory probes and below the Planck cutoff, providing a natural hierarchy compatible with the existing Z-Spin scale structure (ZS-S4 ε-Higgs portal coupling $\xi_{eff} \sim A^2/\left(16\pi^2\right) \approx 4.1 \times 10^{-5}$).

3.4 Infrared Compatibility

The IR behavior of the master action must be compatible with the low-energy Standard Model. The photon, as a linear combination $\gamma = cos\theta_W B_\mu + sin\theta_W W^3_\mu$, remains massless after electroweak symmetry breaking (ZS-U9 NC-U9.7: photon masslessness is taken as experimental input). The GUT-scale Stückelberg mass of B_μ decouples from IR physics at scales $r \gg \lambda_C(B) \sim m_B^{-1} \sim 10^{-33}$ m (see §5.3).

The hypercharge coupling $g_Y \approx 0.345$ used in the Stückelberg mass estimate above is the low-energy (M_Z-scale) value. The running to the compactification scale $m_\rho \approx 0.16 M_P$ or higher is governed by the Standard Model U(1)_Y β-function (ZS-U9 NC-U9.3: ZS-U9 does not derive the SU(5) GUT unification scale or the GUT coupling). A complete RG analysis from M_Z to the Stückelberg scale is outside the scope of this paper; the Stückelberg mass estimate here is an order-of-magnitude statement that establishes the GUT-scale hierarchy without claiming precision on f's numerical realization at the electroweak scale.

[STATUS: CONSISTENT] Stückelberg mass at GUT scale; Standard Model IR behavior preserved.

§4. Theorem S10.2: L_XY = 0 Preservation

The ZS-M6 §7A Continuum Perturbative Protection Theorem [PROVEN-PERTURBATIVE] establishes that $L_{XY}^{eff,\,direct} = 0$ to all orders in perturbation theory for the ZS-F1 action. This section extends the theorem to the ZS-S10 extended action.

4.1 Statement

Theorem S10.2 (L_XY = 0 Preservation, PROVEN-PERTURBATIVE). In the ZS-S10 master action (Definition 3.1), no direct X-sector to Y-sector coupling vertex is generated at any order in perturbation theory:

$$L_{XY}^{eff,\,direct}\Big|_{ZS-S10} = 0 \quad to\ all\ orders\ in\ perturbation\ theory.$$

All X↔Y communication proceeds through the Z-mediator with strength $O\left(\kappa^2\right) = O(A/Q) \approx 0.007$, with higher-loop suppression $(A/4\pi)^{2n}$.

4.2 Proof (Five Steps)

The proof extends the four-step structure of ZS-M6 §7A by adding an independent fifth step from ZS-F2 §4.2A (Schur Protection).

Step 1 (Lorentz Algebra Decomposition — UNCHANGED).

From ZS-M2 §2 [PROVEN]: $so(1,3) \otimes C \cong su(2)_A \oplus su(2)_B$ with $\left[su(2)_A, su(2)_B\right] = 0$. The Z-Spin sector assignment X ↔ su(2)_A (dim 3), Y ↔ su(2)_B (dim 6) inherits this exact commutativity (ZS-F5, PROVEN).

The new terms in the ZS-S10 action are: (a) the gauge-covariant kinetic term $-\frac{1}{2}M_P^2\left|D_\mu\Phi\right|^2$, which is a Z-sector scalar kinetic term with Lorentz-vector gauge coupling, and (b) the standard U(1)_Y gauge kinetic term $-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}$. The gauge field B_μ is a Lorentz 4-vector — a (1/2, 1/2) representation in the (A, B) decomposition — and as an internal U(1)_Y gauge field it is electroweak-sector content, orthogonal to the X, Y, Z polyhedral-sector decomposition.

Therefore the algebraic identity $\left[su(2)_A, su(2)_B\right] = 0$ is **PRESERVED** under the ZS-S10 extension.

Step 2 (Action-Level Absence — EXTENDED).

The ZS-F1 action contains no direct X-Y coupling (ZS-F1 §9, ZS-S1 §4, PROVEN). We verify that the three new ZS-S10 terms preserve this property.

(a) $-\frac{1}{4}B_{\mu\nu}B^{\mu\nu}$*:* Pure U(1)_Y gauge kinetic term. This does not couple X-sector to Y-sector degrees of freedom; B_μ is an internal gauge field independent of the (X, Y, Z) polyhedral sector decomposition.

(b) $-\frac{1}{2}M_P^2 g_Y^2 \kappa^2 Y_f^2 B_\mu B^\mu$ *(Stückelberg mass for B_μ from Step 1 of Theorem S10.1 proof):* Pure gauge sector self-coupling, no X-Y direct coupling.

(c) $M_P^2 \kappa g_Y B_\mu \partial^\mu \theta$ *(mixing term between θ and B_μ):* This is a Z-sector ↔ gauge-sector coupling (θ is the Z-sector Goldstone mode, B_μ is the internal gauge field). It is not an X-Y direct coupling.

Matter-sector fermion couplings: Fermion coupling to B_μ in S_m takes the standard form $g_Y Y_f B_\mu \psi^-_f \gamma^\mu \psi_f$, where Y_f is the individual fermion hypercharge. No fermion bilinear of the form $\psi^-_X \Gamma \psi_Y$ (X-sector fermion to Y-sector fermion direct coupling) is introduced.

Therefore no term in the ZS-S10 action directly couples X-sector to Y-sector degrees of freedom without Z-sector mediation.

Step 3 (Ward-Takahashi Identity — EXTENDED).

Let Q_A^a (a = 1,2,3) be the conserved charges of su(2)_A (X-sector). In any Lorentz-invariant quantum field theory, the Ward-Takahashi identity gives:

$$\partial_\mu \langle T\{J^\mu_{A,a}(x) \cdot O_B\left(y_1\right) \cdots O_B\left(y_n\right)\}\rangle_{connected} = 0$$

where O_B are operators transforming purely under su(2)_B (Y-sector).

In the ZS-S10 extended action, the new Feynman diagram vertices are: $g_Y Y_f B_\mu \psi^-_f \gamma^\mu \psi_f$ (fermion-B coupling), $M_P^2 \kappa g_Y B_\mu \partial^\mu \theta$ (θ-B mixing), and $M_P^2 \kappa^2 g_Y^2 B_\mu B^\mu$ (B self-mass). Any n-loop diagram with external legs attached only to pure X-sector and pure Y-sector operators, and containing no Z-sector propagator, must route through either the B_μ propagator or through Z-sector vertices.

The critical observation: **the θ-B mixing vertex** $M_P^2 \kappa g_Y B_\mu \partial^\mu \theta$ **carries coefficient** $\kappa = \sqrt{A/Q}$ — the Z-sector per-mode coupling strength established by ZS-M6 Theorem 2.2.1. Therefore any path that connects X to Y through B_μ must insert at least one factor of κ (through the θ-B mixing vertex when θ is coupled at internal lines), making the contribution $O(\kappa^2)$ minimum at tree level and higher-order suppressed at loops.

Equivalently, fermion-B-fermion chains: the fermion-B vertex $g_Y Y_f B_\mu \psi^-_f \gamma^\mu \psi_f$ couples X-sector and Y-sector fermions to a common internal B_μ, but this coupling channel is precisely the Z-mediated indirect channel (via the Stückelberg-screened B_μ propagator at scales below m_B). In the effective theory below m_B, the B_μ mediation is integrated out and contributes at $O(\kappa^2)$ to effective X-Y transitions, exactly as before.

Therefore the Ward-Takahashi identity is preserved; only Z-mediated indirect coupling of magnitude $O(\kappa^2)$ is permitted.

Step 4 (Anomaly-Free Verification — EXTENDED).

The original anomaly-free verification of ZS-M6 §7A Step 4 covers (4a) scalar Φ has no chirality, (4b) SM ABJ anomalies act on gauge sector only, (4c) no mixed $su(2)_A \times su(2)_B$ anomaly.

Four additional considerations for ZS-S10:

(4d) Stückelberg chiral anomaly: The U(1)_Y Stückelberg coupling could potentially generate chiral anomalies. However, ZS-U9 §7 [PROVEN at integer precision] establishes that all five Standard Model anomaly cancellation conditions A1-A5 PASS for the Trinity Braiding hypercharges:

$$A_1: [SU(3)]^2 \cdot U(1)_Y: \sum_{quarks} \left(2Y_Q - Y_u - Y_d\right) = 0 ✓$$

$$A_2: [SU(2)]^2 \cdot U(1)_Y: 3Y_Q + Y_L = 0 ✓$$

$$A_3: \left[U(1)_Y\right]^3: \sum_f 2Y_f^3 = 0 ✓$$

$$A_4: [gravity]^2 \cdot U(1)_Y: \sum_f Y_f = 0 ✓$$

$$A_5: [SU(2)]^4\ (Witten): integer\ multiple\ of\ 2 ✓$$

Since the Z-bias field Φ is a scalar (spin 0, no chirality), it contributes zero to all five triangle-type or global anomalies regardless of its U(1)_Y charge q_Φ = +1 (Theorem S10.4). The Trinity Braiding anomaly cancellation is preserved.

(4e) Stückelberg gauge invariance: Under $B_\mu \to B_\mu + \partial_\mu \Lambda$, $\theta \to \theta + \kappa g_Y \Lambda$, the mixing combination $\left(\partial_\mu \theta - \kappa g_Y B_\mu\right)$ is invariant. This gauge symmetry acts only on the internal U(1)_Y sector and does not affect the Lorentz decomposition or the (X, Y, Z) polyhedral sectors.

(4f) Mixed Lorentz × U(1)_Y anomaly: Since the gravitational-U(1)_Y triangle anomaly $\sum_f Y_f = 0$ (ZS-U9 condition A4), both $su(2)_A \times U(1)_Y$ and $su(2)_B \times U(1)_Y$ mixed anomalies cancel individually.

(4g) Φ's contribution to U(1)_Y triangle anomalies: As a scalar field, Φ does not contribute to any triangle anomaly (fermion-loop triangle anomalies require chiral fermions). Therefore its U(1)_Y charge $q_\Phi = +1$ does not enter any anomaly condition.

Therefore the Ward identity of Step 3 is anomaly-free in the ZS-S10 extended action.

Step 5 (Schur Protection Independent Layer — PRESERVED).

From ZS-F2 §4.2A (Adjoint Obstruction Theorem, PROVEN): A_5 is the unique finite subgroup of SO(3) for which $adj(SU(3))|_\Gamma = 3 \oplus 5$ lacks the irrep 3′. By Schur's Lemma, no A_5-equivariant intertwiner exists between the irreps 3 ($\chi_3(C_5) = \varphi$) and 3′ ($\chi_{3'}(C_5) = 1 - \varphi$), where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio.

This discrete representation-theoretic protection acts on the SM gauge sector (SU(3)_C × SU(2)_L) assignment to I-irreps, which is orthogonal to the U(1)_Y Stückelberg extension introduced in ZS-S10. The Adjoint Obstruction Theorem is therefore unaffected by ZS-S10.

4.3 Conclusion and Scope

Combining Steps 1-5: $L_{XY}^{eff,\,direct} = 0$ to all orders in perturbation theory in the ZS-S10 action.

Scope and Limitations. This theorem covers:

- All perturbative orders in Lorentz-invariant regularization
- The weak-curvature regime ($R \ll M_P^2$)

It does not cover:

- Non-perturbative effects (instantons are expected to be single-sector objects per ZS-M2 §6.1, but a rigorous proof is absent)
- Strong-curvature regime ($R \sim M_P^2$) where the perturbative expansion breaks down (ZS-A3 §7 Sector Duality is HYPOTHESIS)

[STATUS: PROVEN-PERTURBATIVE] Inherited from ZS-M6 §7A with ZS-S10 extensions verified in Steps 2, 3, 4 and independent Schur layer Step 5.

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§5. Theorem S10.3: Stückelberg-Corollary IV Bridge

This section establishes that the ZS-S10 master action realizes both branches of ZS-A7 Corollary IV (Vortex Bose/Fermi Duality) — the inner (F) j=1/2 spinor branch and the outer (B) 2π Goldstone winding branch — as two limits of a single action-level object.

5.1 Statement

Theorem S10.3 (Stückelberg-Corollary IV Bridge, DERIVED). The ZS-S10 master action simultaneously realizes:

(i) UV limit (vortex core, r ~ ℓ_P): The Z-Anchor boundary condition $|\Phi(x_0)| = 0$ (ZS-F1 §5.2 PROVEN, ZS-A6 §4.5.6 cigar bounce DERIVED) together with the gauge-covariant derivative $D_\mu \Phi$

realizes the **Corollary IV (F) branch**: a j = 1/2 spinor carrier with 4π closure period. Explicit action-level proof: Theorem S10.5-BPS (§7).

(ii) IR limit (galactic scale, r ≫ λ_C(B) ~ m_B⁻¹): Exponential Stückelberg screening of the B_μ gauge boson ($e^{-m_B r}$) reduces the gauge-covariant derivative $D_\mu\theta = \left(\partial_\mu\theta - \kappa g_Y B_\mu\right) \to \partial_\mu\theta$, and the Goldstone mode satisfies $\Box\theta = 0$ with integer winding $\oint_C d\theta = 2\pi n$ (π_1(U(1)) = ℤ, PROVEN). This realizes the **Corollary IV (B) branch**.

(iii) Common boundary condition: Both limits share the Z-Anchor boundary condition $\left|\Phi\left(x_0\right)\right| = 0$ at the vortex core (ZS-F1 §5.2, PROVEN with horizon realization DERIVED via ZS-A6 §4.5.6).

5.2 B_μ = 0 Limit: ZS-F1 Recovery

Setting $B_\mu \equiv 0$ in the ZS-S10 master action gives:

$$S_{S10}|_{B_\mu=0} = \int d^4x\sqrt{-g}\left[\frac{1}{2}M_P^2\left(1 + A|\Phi|^2\right)R - \frac{1}{2}M_P^2|\partial\Phi|^2 - V(\Phi)\right] + S_m|_{B_\mu=0}$$
$$= S_{F1}$$

exactly. All ZS-F1 equations of motion, vortex solutions, Goldstone halo dynamics, and inflationary background cosmology are recovered identically. This is the **kinematic backward compatibility** statement: ZS-S10 contains ZS-F1 as a gauge-fixed limit ($B_\mu = 0$ corresponds to a specific choice of U(1)_Y gauge).

5.3 Galactic Scale: ZS-A1 78/78 PASS Preservation

At galactic scales, the Stückelberg mass of B_μ is $m_B = \kappa g_Y M_P \approx 0.029 M_P \approx 7.1 \times 10^{16}$ GeV (Theorem S10.1), yielding Compton wavelength:

$$\lambda_C(B) = \hbar/\left(m_B c\right) \approx 1/\left(7.1 \times 10^{16}\, GeV\right) \approx 2.8 \times 10^{-33}\, m$$

For galactic scales $r \sim 10$ kpc $= 3.09 \times 10^{20}$ m:

$$\frac{r}{\lambda_C(B)} \approx \frac{3.09\times10^{20}}{2.8\times10^{-33}} \approx 10^{53}$$

The B_μ field is exponentially screened at galactic scales: $B_\mu \sim e^{-m_B r} \to 0$ with suppression factor $e^{-10^{53}}$, which is effectively zero at any physical precision. The gauge-covariant derivative reduces to:

$$D_\mu\theta|_{r\gg\lambda_C(B)} = \partial_\mu\theta - \kappa g_Y B_\mu \to \partial_\mu\theta$$

Consequence: The Goldstone equation of motion $\Box\theta = 0$ (ZS-A1 §2.1 DERIVED) is preserved at galactic scales. The logarithmic profile $\theta(r) = ln\left(r/r_s\right)/L$ and isothermal halo density $\rho_\theta = M_P^2/\left(2L^2r^2\right)$ (ZS-A1 §2.2 DERIVED) are **unchanged**. All 78 ZS-A1 verification tests PASS at full precision.

[STATUS: DERIVED] Galactic-scale Goldstone physics preserved at precision of $O\left(e^{-10^{53}}\right)$, i.e., absolutely.

5.4 Inflation: ZS-U1 r = 0.0089 Preservation

At inflationary scales, $|\Phi| \gg 1$ (ZS-U1 §4, DERIVED). The inflaton is identified as the radial mode $\rho = |\Phi|$, and the angular mode θ is frozen at the Hubble scale (Lyth-Riotto 1999, isocurvature suppression). In this regime:

$$D_\mu \Phi = \left(\partial_\mu - i\kappa g_Y B_\mu\right)\rho\, e^{i\theta}$$

with $\partial_\mu \theta \approx 0$. The inflationary Lagrangian density includes:

$$-\tfrac{1}{2}M_P^2\left|D_\mu \Phi\right|^2\Big|_{inflation} \approx -\tfrac{1}{2}M_P^2\left(\partial_\mu \rho\right)^2 - \tfrac{1}{2}M_P^2\rho^2\kappa^2 g_Y^2 B_\mu B^\mu$$

The Stückelberg mass correction to the inflaton potential is $\Delta V \sim \rho^2\kappa^2 g_Y^2 M_P^2 B^2$, which vanishes for $B_\mu \to 0$ in the inflationary vacuum. The slow-roll parameters ϵ_V, η_V computed from $V(\Phi) = (\lambda/4)M^4_P(|\Phi|^2-1)^2$ (ZS-U1 §4.2 DERIVED) are **unchanged**.

Consequence: Tensor-to-scalar ratio $r = 0.0089$ (ZS-U1 prediction, DERIVED) is preserved. Spectral tilt n_s is unchanged. CMB scalar amplitude $A_s = 2.1 \times 10^{-9}$ (Planck 2018 input) is unchanged.

[STATUS: DERIVED] Inflation observables preserved through direct inspection of the Lagrangian structure in the $|\Phi| \gg 1$ regime.

5.5 Vortex Core: Corollary IV (F) Realization

At the vortex core (Region I of ZS-F1 §5.3, $r \sim \xi \sim \ell_P$), $|\Phi| \to 0$ by Z-Anchor (ZS-F1 §5.2 PROVEN, ZS-A6 §4.5.6 cigar bounce realization DERIVED at 0.089% Wick-rotation match precision).

The gauge-covariant kinetic term at the vortex core:

$$\left|D_\mu \Phi\right|^2\Big|_{|\Phi|\to 0} = \left(\partial_\mu \rho\right)^2 + \rho^2\left(\partial_\mu \theta - \kappa g_Y B_\mu\right)^2 \to \left(\partial_\mu \rho\right)^2 + O\left(\rho^2\right)$$

The gauge coupling through the θ-mode vanishes as $\rho \to 0$, but the **phase structure remains non-trivial** through the topological winding number n around the vortex. Specifically, the fluctuation $\Phi \approx c_1(x + iy)$ near the core (Step 2 of Theorem S10.5-BPS proof in §7.3) realizes the j = 1/2 spinor structure of ZS-M3 Theorem 5.1.

Consequence: ZS-A7 Corollary IV (F) branch — the j = 1/2 spinor at the vortex core — is realized at the action level through the ZS-S10 master action. Explicit proof with Kraus-operator 4π closure: Theorem S10.5-BPS (§7).

5.6 Galactic Halo: Corollary IV (B) Realization

At galactic scales (Region II of ZS-F1 §5.3, $r_s \ll r \ll r_Z$), $|\Phi| \approx 1$ (radial mode frozen by $m_\rho \sim M_P$) and only the Goldstone θ varies. With B_μ exponentially screened (§5.3), the winding integral:

$$\oint_C D_\mu \theta\, dx^\mu = \oint_C \left(\partial_\mu \theta - \kappa g_Y B_\mu\right) dx^\mu \to \oint_C \partial_\mu \theta\, dx^\mu = 2\pi n$$

where the last equality uses $\pi_1(U(1)) = Z$ (ZS-A6 §4.4.2, PROVEN) and the Stückelberg screening of B_μ at galactic scales (§5.3).

Consequence: ZS-A7 Corollary IV (B) branch — the 2π Goldstone winding — is realized at the action level in the IR limit of the ZS-S10 master action.

5.7 Summary of Corollary IV Action-Level Realization

Table 5.1. Mapping of ZS-S10 regimes onto Corollary IV branches.

Regime	Radius scale	Field configuration	Corollary IV branch	Observable
UV (core)	r ~ ξ ~ ℓ_P	$\lvert\Phi\rvert \to 0$, Frobenius $f \sim c_1\rho$	(F) j=1/2 spinor, 4π closure	$\tilde{u}_{seam}(\alpha) = cos(\alpha/2)$
Intermediate	r ~ λ_C(B) ~ 10^{-33} m	$\lvert\Phi\rvert \sim 1$, B_μ transitioning	Crossover (neither pure F nor pure B)	n/a
IR (galactic)	r ≫ λ_C(B)	$\lvert\Phi\rvert \approx 1$, B_μ screened	(B) 2π Goldstone winding	$\oint d\theta = 2\pi n$

The two branches coexist on the same vortex line, sharing the common boundary condition $\left|\Phi\left(x_0\right)\right| = 0$.

This is the exact content of ZS-A7 Corollary IV (§4.4.2), now realized at the action level.

[STATUS: DERIVED] Theorem S10.3 established from existing PROVEN/DERIVED inputs: ZS-F1 §3.1 (action), ZS-F1 §5.2 (Z-Anchor), ZS-M6 Thm 2.2.1 ($\kappa^2 = A/Q$), ZS-A6 §4.5.6 (cigar bounce), ZS-A7 §4.4 (Corollary IV), ZS-A1 §2 (Goldstone halo), ZS-U1 §4 (inflation). Zero new parameters.

§6. Theorem S10.4: Φ Field U(1)_Y Charge

This section establishes the U(1)_Y charge of the Z-bias field Φ through four independent constraints, three of which are independent of ZS-S10 itself, thereby removing the apparent circularity of the Gap G1 self-closure.

6.1 Statement

Theorem S10.4 (Φ Field U(1)_Y Charge, DERIVED). The U(1)_Y charge of the Z-bias field Φ in the ZS-S10 master action is uniquely fixed to:

$$q_\Phi = +1$$

6.2 The Four Independent Paths

The theorem is established through four independent constraints, the first three of which are independent of the ZS-S10 definition.

Path I — ZS-F1 §3.2 Fundamental Convention [PROVEN, corpus-LOCKED].

By the explicit statement of ZS-F1 §3.2 [PROVEN, direct quotation]: "The action is invariant under $\Phi \to \exp(i\alpha)\Phi$ for constant $\alpha \in [0, 2\pi)$."

This statement **definitionally** establishes Φ as a charge-(+1) character of U(1)_Z. The gauge transformation parameter α enters with coefficient 1 in the exponent (not 2, 3, or −1), and this is a **corpus-LOCKED convention** of the ZS-F1 action. Any other choice of q^(Z)_Φ would contradict the direct statement of ZS-F1 §3.2.

$$c_\Phi^{(Z)} = +1 \quad [PROVEN\ by\ corpus\ definition]$$

Path II — ZS-U9 Theorem 3.1 Integer Quantization [PROVEN].

By ZS-U9 Theorem 3.1 [PROVEN]: For any state ψ transforming as $\psi \to e^{ic\alpha}\psi$ under U(1)_Z with $\alpha \in [0, 2\pi)$, single-valuedness $\psi(\alpha + 2\pi) = \psi(\alpha)$ forces $c \in Z$.

Combined with Path I, this gives $c_\Phi^{(Z)} = +1 \in Z$ uniquely and consistently.

Path III — Vortex Winding Number Observational Selection [DERIVED].

By ZS-A6 §4.4.1 [DERIVED] and §4.5.6 [DERIVED, 2026-04 cigar bounce closure with 0.089% Wick-rotation match], the quantitatively-verified Z-Anchor profile uses $n = 1$ winding. The Frobenius exponent $\alpha_{Frob} = n/2 = 1/2$ is confirmed numerically.

While Path III does not by itself prove that internal U(1)_Z charge $c_\Phi^{(Z)}$ equals the spatial winding n (ZS-U9 §3.3 states only that "both external topological winding and internal representation charge respect integer quantization"), the **supporting evidence** converges on $c_\Phi^{(Z)} = 1$ through the **minimal coupling ansatz** of ZS-S10: a gauge-covariant derivative $D_\mu\Phi = \left(\partial_\mu - iq_\Phi \kappa g_Y B_\mu\right)\Phi$ with $q_\Phi \neq c_\Phi^{(Z)}$ would require a compensating rescaling of the action, contradicting the LOCKED normalization of ZS-F1.

[STATUS: Consistent with $c_\Phi^{(Z)} = 1$; not an independent proof; serves as supporting evidence.]

Path IV — EFT Kinematic Bound [CONSISTENT].

Under the ZS-S10 master action, the Stückelberg mass of B_μ coupled to Φ with charge q_Φ is $m_B = q_\Phi \kappa g_Y M_P$. The requirement that the Stückelberg mass remains below the Planck cutoff ($m_B \leq M_P$) for EFT validity gives:

$$q_\Phi \leq \frac{1}{\kappa g_Y} \approx \frac{1}{0.0853 \times 0.345} \approx 34$$

Combined with the integer quantization of Path II, this gives $1 \leq q_\Phi \leq 34$, consistent with $q_\Phi =+ 1$ but not uniquely selecting it. An additional kinematic constraint — the Z-Spin register cap $Q = 11$ — narrows this to $1 \leq q_\Phi \leq 11$, still consistent with $q_\Phi =+ 1$.

6.3 Complete Proof

Step 1 [PROVEN, corpus-LOCKED]: Path I establishes $c_\Phi^{(Z)} =+ 1$ by the ZS-F1 §3.2 definitional convention. This is **independent of ZS-S10**.

Step 2 [PROVEN]: Path II confirms integer-quantization consistency.

Step 3 [DERIVED, observational]: Path III provides supporting evidence from $n = 1$ vortex profile.

Step 4 [DERIVED by Gap G1 closure]: Under the ZS-S10 Stückelberg minimal coupling $D_\mu\Phi = \left(\partial_\mu - iq_\Phi \kappa g_Y B_\mu\right)\Phi$, the U(1)_Z charge of Φ transfers to its U(1)_Y charge: $q_\Phi^{(Y)} = c_\Phi^{(Z)} =+ 1$. This identification is the **definitional statement of Gap G1 closure**, not a circular derivation.

Step 5 [PROVEN, ZS-U9 §7]: Anomaly cancellation A1-A5 are preserved unchanged, since the Z-bias field Φ is a scalar (no chiral contribution to triangle anomalies) for any integer q_Φ.

Therefore $q_\Phi =+ 1$ is uniquely fixed by Paths I-II (corpus-LOCKED and PROVEN), supported by Path III (observational), bounded by Path IV (EFT consistency), and identified with $q_\Phi^{(Y)}$ via Step 4 (definitional Gap G1 closure).

6.4 Circularity Resolution

The apparent circularity of the original Theorem S10.4 — "Gap G1 closure is used to derive q_Φ, but Gap G1 closure is itself defined by ZS-S10" — is **resolved** as follows:

- **Paths I, II are independent of ZS-S10**: they fix $c_\Phi^{(Z)} =+ 1$ from the ZS-F1 §3.2 corpus-LOCKED convention and ZS-U9 Theorem 3.1 PROVEN integer quantization.

- **Path III provides observational support independent of ZS-S10**: the $n = 1$ vortex profile is already established in ZS-A6 at 0.089% precision.
- **Path IV is a kinematic consistency check**: the EFT validity range $1 \leq q_\Phi \leq Q = 11$ is consistent with $q_\Phi =+ 1$.
- **Step 4 (U(1)_Z → U(1)_Y)** is the **definitional statement** of ZS-S10's minimal coupling, not a derived identity subject to circular reasoning. The identification is precisely what ZS-S10 provides at the action level as the closure of Gap G1.

This is the same self-consistent bootstrapping pattern as the Trinity Braiding Theorem (ZS-U9 §6), where each ingredient's individual gap is closed by another ingredient's capability. In ZS-S10's case, $c_\Phi^{(Z)}$ is determined externally (Path I-III), and the ZS-S10 action then identifies this externally-determined charge with the U(1)_Y charge by definition.

[STATUS: DERIVED] (upgraded from DERIVED-CONDITIONAL in the preliminary version). The circularity is resolved because Paths I-III fix $c_\Phi^{(Z)} =+ 1$ outside ZS-S10, and the U(1)_Z ↔ U(1)_Y identification is definitional rather than derivational.

6.5 Consequences for Stückelberg Mass Spectrum

With $q_\Phi =+ 1$:

$$m_B = q_\Phi \cdot \kappa \cdot g_Y \cdot M_P = 1 \cdot \sqrt{A/Q} \cdot g_Y \cdot M_P$$

Numerically: $m_B \approx 0.0853 \times 0.345 \times 2.435 \times 10^{18}$ GeV $\approx 7.1 \times 10^{16}$ GeV.

For lepton doublet (Y_L = −1/2), the fermion-sector Stückelberg mass (contribution to effective gauge-fermion coupling mass scale):

$$m_B^{(L)} = g_Y \left|Y_L\right| f = 0.345 \times 0.5 \times 2.078 \times 10^{17} = 3.58 \times 10^{16}\ GeV$$

For electron (Y_e = −1):

$$m_B^{(e)} = g_Y \left|Y_e\right| f = 0.345 \times 1 \times 2.078 \times 10^{17} = 7.17 \times 10^{16}\ GeV$$

These are GUT-scale masses, safely above all laboratory probes.

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§7. Theorem S10.5-BPS: Bogomolnyi BPS Spinor Lift

This section provides the seven-step explicit action-level proof that the ZS-A7 Theorem 3.2-bis Kraus-operator 4π closure emerges naturally from the action content of ZS-S10 at the Z-anchored vortex core.

7.1 Statement

Theorem S10.5-BPS (Bogomolnyi BPS Spinor Lift, DERIVED). Under the ZS-S10 master action (Definition 3.1) with a Z-anchored vortex solution $\Phi(\rho, \phi) = f(\rho)e^{in\phi}$ satisfying the cigar vortex equation (ZS-A6 §4.5.6, DERIVED):

(i) The vortex solution with $n = 1$ and boundary conditions $f(0) = 0$, $f(\infty) = 1$ exists uniquely (Jaffe-Taubes 1980, PROVEN).

(ii) Near the vortex core $\rho \to 0$, linearization gives $\Phi \approx c_1(x + iy)$, identifying the Z-sector 2-dimensional real subspace $\vec{\Psi} = (Re\,\Phi, Im\,\Phi)^T \in R^2$.

(iii) The complex structure $J: R^2 \to R^2$ satisfies $J = -i\sigma_y$, where σ_y is the Pauli y-matrix.

(iv) Under U(1)_Z gauge rotation $\Phi \to e^{i\alpha}\Phi$, the vector $\vec{\Psi}$ transforms by $R(\alpha) = exp\left(-i\alpha\sigma_y\right)$ (vector / j = 1 rep).

(v) The Kraus operators $\{K_0, K_1\}$ in the Z-mediated CPTP channel transform by half-angle: $U_Z(\alpha) = exp\left(-i\alpha\sigma_y/2\right)$ (spinor / j = 1/2 rep).

(vi) By ZS-M3 Lemma 10.1 ($D^{1/2}(-I) = -I$): $K_z(\alpha + 2\pi) = -K_z(\alpha)$, $K_z(\alpha + 4\pi) = +K_z(\alpha)$.

(vii) By ZS-M3 Theorem 5.1 (j = 1/2 uniqueness on dim = 2 intertwiner), this spinor realization is unique.

7.2 Step 1: Vortex Existence

By Jaffe-Taubes (1980) and ZS-A6 §4.5.3 [PROVEN, Theorem C3], the cigar vortex equation on the near-horizon cigar metric:

$$f''(\rho) + \frac{1}{\rho}f'(\rho) - \frac{f}{\rho^2} - \lambda^* f\left(f^2 - 1\right) = 0$$

with $\lambda^* = 2A^2 = 0.01283$ [ZS-U5 §15.4 DERIVED-CONDITIONAL], has a **unique smooth solution** in the $n = 1$ winding sector with boundary conditions $f(0) = 0$ (Z-Anchor) and $f(\rho \to \infty) = 1$ (vacuum). Existence is proven via the direct variational method; uniqueness in the $n = 1$ sector is proven via the Weinberg-Salam convexity argument (Jaffe-Taubes Theorem C3).

Numerical realization (ZS-A6 §4.5.6.2): $c_{cigar} = f'(0) = 0.06605585$, verified via scipy.solve_bvp at rms residual 7.31×10^{-11} across 1632 mesh nodes.

7.3 Step 2: Linear Core Behavior

Near the vortex core $\rho \to 0$, the Frobenius expansion of the cigar vortex equation yields:

$$f(\rho) = c_1\rho + c_3\rho^3 + O\left(\rho^5\right)$$

with $c_1 \approx 0.06606$ (cigar coordinates) or equivalently $c_1 \approx 0.935$ in EF-BVP coordinates (0.089% Wick-rotation match, ZS-A6 §4.5.6.4). Substituting into the vortex ansatz $\Phi = f(\rho)e^{in\phi}$ with $n = 1$:

$$\Phi(\rho \to 0) = c_1\rho(cos\phi + isin\phi) = c_1\rho cos\phi + ic_1\rho sin\phi = c_1(x + iy)$$

using $x = \rho cos\phi$ and $y = \rho sin\phi$.

Thus the Z-bias field Φ linearizes to $c_1(x + iy)$ at the vortex core, a complex-linear function of spatial coordinates. Decomposing into real and imaginary parts:

$$\vec{\Psi}(x, y) \equiv (Re\,\Phi\ Im\,\Phi) = c_1(x\ y) \in R^2$$

This 2-dimensional real vector space is the Z-sector carrier (ZS-F1 §2.3 DERIVED: dim(Z) = 2 identifies Z-sector state space with $\mathbb{C}$).

7.4 Step 3: Complex Structure → Pauli Matrix

The isomorphism $C \cong R^2$ via $z = x + iy \leftrightarrow (x, y)^T$ equips R^2 with a natural complex structure: multiplication by i on C corresponds to 90° rotation on R^2:

$$i(x + iy) = -y + ix \leftrightarrow (-y\ x) = J(x\ y), \quad J = (0\ -1\ 1\ 0)$$

Direct comparison with the Pauli y-matrix σ_y:

$$\sigma_y = (0\ -i\ i\ 0), \quad -i\sigma_y = (0\ -1\ 1\ 0) = J$$

Therefore $J = -i\sigma_y$, an algebraic identity relating the complex structure of $C \cong R^2$ to the Pauli y-matrix of SU(2). The $su(2)$ algebra action on the Z-sector state space is naturally realized through σ_y. ▫3

7.5 Step 4: Vector Rotation (j = 1 representation)

Under U(1)_Z gauge rotation $\Phi \to e^{i\alpha}\Phi$, the complex field transforms as:

$$e^{i\alpha}(x + iy) = (cos\alpha + isin\alpha)(x + iy) = (xcos\alpha - ysin\alpha) + i(xsin\alpha + ycos\alpha)$$

In the real vector representation:

$$\vec{\Psi} \to R(\alpha)\vec{\Psi} = (cos\alpha\ -sin\alpha\ sin\alpha\ cos\alpha)\vec{\Psi}$$

The matrix $R(\alpha)$ is the matrix exponential of αJ:

$$R(\alpha) = exp(\alpha J) = cos\alpha \cdot I + sin\alpha \cdot J = cos\alpha \cdot I - isin\alpha \cdot \sigma_y = exp\left(-i\alpha\sigma_y\right)$$

This is the **j = 1 (vector) representation** of SU(2) on the 2-dimensional real space (equivalent to the spin-1 spherical tensor rep restricted to the plane). $R(\alpha + 2\pi) = R(\alpha)$, so the vector rep is 2π-periodic.

7.6 Step 5: Kraus Square Root (j = 1/2 representation)

From ZS-Q1 §3.3 [PROVEN], the Z-mediated CPTP channel admits a Kraus decomposition $\Lambda\left(\rho_X\right) = \sum_z K_z \rho_X K_z^\dagger$ with Kraus count = dim(Z) = 2 (ZS-A4 Appendix A). The Kraus operators are **wave-function-linear objects** (not density-matrix-bilinear), as emphasized by ZS-A7 Theorem 3.2-bis §3.2-bis.3.

Key quantum-mechanical principle: For a CPTP channel Λ with Choi state $C_\Lambda = \sum_z \left|K_z\right\rangle\rangle\langle\langle K_z\right|$ transforming under a U(1) gauge rotation by $R(\alpha) = exp\left(-i\alpha\sigma_y\right)$ at the **bilinear** level, the Kraus operators themselves transform at the **linear** level via the square root:

$$K_z(\alpha) = \left[\sqrt{R(\alpha)}\right]_{zz'} K_{z'}(0) = \left[exp\left(-i\alpha\sigma_y/2\right)\right]_{zz'} K_{z'}(0)$$

Setting $U_Z(\alpha) = exp\left(-i\alpha\sigma_y/2\right)$, this is the **j = 1/2 (spinor) representation** of SU(2).

Consistency with Choi state 2π periodicity [ZS-A7 Theorem 3.2-bis Part 2]: The Choi state $C_\Lambda(\alpha) = \sum_z K_z(\alpha) \otimes K_z^*(\alpha) = \sum_z \left[U_Z(\alpha)K(0)\right] \otimes \left[U_Z(\alpha)K(0)\right]^*$ is unitarily-invariant under $U_Z(\alpha)$ (bilinear structure), hence 2π-periodic (the full U(1) rotation leaves it invariant). This matches ZS-A7 Theorem 3.2-bis Part 2 exactly.

7.7 Step 6: 4π Closure (ZS-M3 Lemma 10.1)

By direct Pauli-matrix algebra:

$$exp\left(-i\pi\sigma_y\right) = cos\pi \cdot I - isin\pi \cdot \sigma_y = -I$$

(standard SU(2) center element, equivalent to ZS-M3 Lemma 10.1: $D^{1/2}(-I) = -I$, PROVEN). Therefore:

$$U_Z(\alpha + 2\pi) = exp\left(-\, i(\alpha + 2\pi)\sigma_y/2\right) = exp\left(-\, i\alpha\sigma_y/2\right) \cdot exp\left(-\, i\pi\sigma_y\right) = U_Z(\alpha) \cdot (-\, I) =- U_Z(\alpha)$$

Applied to Kraus operators:

$$K_z(\alpha + 2\pi) =- K_z(\alpha), \quad K_z(\alpha + 4\pi) = (-\, 1)^2 K_z(\alpha) =+ K_z(\alpha)$$

This is the **4π closure at the Kraus-operator level**, matching ZS-A7 Theorem 3.2-bis Part 3 exactly.

7.8 Step 7: Uniqueness (ZS-M3 Theorem 5.1)

By ZS-M3 Theorem 5.1 [PROVEN]: among all half-integer spins j, $dim\left(Inv_4(j)\right) = 2$ if and only if $j = 1/2$.

The Z-sector has dim = 2 (ZS-F5, PROVEN), so only j = 1/2 is a viable representation. No alternative spin representation (j = 1, 3/2, 5/2, ...) is possible for dim(Z) = 2.

Therefore the j = 1/2 spinor realization at the vortex core is **unique** — the only possible representation compatible with the Z-sector dimension.

7.9 Observable Realization: Signed Seam Witness

The signed seam witness $\tilde{u}_{seam}(\alpha)$ of ZS-A7 Theorem 3.2-bis Part 3 has an explicit form in terms of the spinor rotation $U_Z(\alpha)$. With orthonormal Kraus basis $Tr\left(K_z^\dagger K_{z'}\right) = \delta_{zz'} \parallel K_0(0)\parallel_F^2$ (standard Kraus normalization, ZS-Q1 §3.3):

$$\tilde{u}_{seam}(\alpha) = \frac{Re\, Tr\left(K_0(0)^\dagger K_0(\alpha)\right)}{\parallel K_0(0)\parallel_F^2}$$

Computing $K_0(\alpha) = \left[U_Z(\alpha)\right]_{00} K_0(0) + \left[U_Z(\alpha)\right]_{01} K_1(0)$:

$$U_Z(\alpha) = (cos(\alpha/2) \;\; - \, sin(\alpha/2) \; sin(\alpha/2) \; cos(\alpha/2)\,)$$

Therefore:

$$\tilde{u}_{seam}(\alpha) = Re\,[cos(\alpha/2) \cdot 1 \, - \, sin(\alpha/2) \cdot 0] = cos(\alpha/2)$$

Explicit 4π periodicity verification:

α	α/2	$\tilde{u}_{seam}(\alpha) = cos(\alpha/2)$
0	0	+ 1
π	π/2	0
2π	π	− 1 (sign flip)
3π	3π/2	0
4π	2π	+ 1 (return to identity)

$$\tilde{u}_{seam}(\alpha + 2\pi) = cos(\alpha/2 + \pi) =- cos(\alpha/2) =- \tilde{u}_{seam}(\alpha) ✓$$

$$\tilde{u}_{seam}(\alpha + 4\pi) = cos(\alpha/2 + 2\pi) =+ cos(\alpha/2) =+ \tilde{u}_{seam}(\alpha) ✓$$

7.10 Theorem Conclusion

All 7 steps follow from PROVEN/DERIVED corpus inputs:

- Step 1: Jaffe-Taubes 1980 + ZS-A6 §4.5.3 Theorem C3 (PROVEN)
- Step 2: ZS-A6 §4.5.6 Frobenius expansion (DERIVED, verified at 0.089%)
- Step 3: $J =- i\sigma_y$ (algebraic identity, PROVEN)

- Step 4: Matrix exponential + trigonometry (PROVEN)
- Step 5: Quantum-mechanical amplitude-wavefunction relation + ZS-A7 Theorem 3.2-bis (PROVEN)
- Step 6: ZS-M3 Lemma 10.1 (PROVEN) + Pauli algebra
- Step 7: ZS-M3 Theorem 5.1 (PROVEN, uniqueness)

Zero new parameters, zero new fields, zero new postulates.

[STATUS: DERIVED] Action-level realization of ZS-A7 Corollary IV (F) branch with explicit observable $\tilde{u}_{seam}(\alpha) = cos(\alpha/2)$ matching F-A7.3 experimental prediction (TESTABLE on Z-Spin quantum hardware 2026-2028).

§8. Non-Claims

Consistent with Z-Spin Collaboration methodology, we enumerate explicit non-claims for ZS-S10:

NC-S10.1. ZS-S10 does NOT introduce a new gauge field beyond the Standard Model hypercharge gauge field B_μ. The ZS-S10 master action identifies U(1)_Z (of ZS-F1 §3.2) with U(1)_Y (of the Standard Model) at the action level; no separate gauge field is postulated.

NC-S10.2. ZS-S10 does NOT derive photon masslessness from first principles. The condition that the photon $\gamma = cos\theta_W B_\mu + sin\theta_W W^3_\mu$ remains massless after electroweak symmetry breaking is taken as experimental input ($m_\gamma < 10^{-18}$ eV PDG 2024), identical to ZS-U9 NC-U9.7. The Stückelberg mechanism of ZS-S10 generates a GUT-scale Stückelberg mass for B_μ itself, not for the physical photon after diagonalization.

NC-S10.3. ZS-S10 does NOT modify any galactic-scale prediction of ZS-A1 (rotation curves, BTFR, M-σ relation, vortex glass profile). Stückelberg screening at galactic scales is exponential with suppression factor $e^{-10^{53}}$, preserving all 78 ZS-A1 verification tests at full precision. See §5.3.

NC-S10.4. ZS-S10 does NOT modify any inflationary prediction of ZS-U1. The tensor-to-scalar ratio $r = 0.0089$, spectral tilt n_s, and CMB scalar amplitude A_s are all preserved unchanged in the $|\Phi| \gg 1$ inflationary regime. See §5.4.

NC-S10.5. ZS-S10 does NOT re-derive the Higgs VEV $v = 245.93$ GeV (ZS-S4 §6.12 DERIVED), the Weinberg angle $sin^2\theta_W = (48/91)x^* = 0.23118$ (ZS-S1 §8.2 PROVEN), or the strong coupling $\alpha_s = 11/93 = 0.11828$ (ZS-S1 §8.1 DERIVED). These upstream results are preserved as input constraints.

NC-S10.6. ZS-S10 does NOT close Gap G2 of the Trinity Braiding Theorem (ZS-M9 Table 2 assignment upgrade from HYPOTHESIS strong to PROVEN). While Theorem S10.2's anomaly preservation is consistent with the Table 2 assignments, a rigorous representation-theoretic proof that the Table 2 assignments are uniquely forced remains OPEN. See NC-U9.2.

NC-S10.7. ZS-S10 does NOT introduce supersymmetry or any new symmetry beyond U(1)_Y Stückelberg. The vortex Bose/Fermi duality of Corollary IV is a topological statement (j=1/2 SU(2) double-cover on Z-sector vs π_1(U(1))=ℤ on vacuum manifold); no SUSY superpartner pairing is invoked (inherited from ZS-A7 NC-A7.8).

NC-S10.8. ZS-S10 does NOT provide a dynamical selection mechanism for $|Q_e| = 1$ at the action level beyond the minimal-integer-charge argument of ZS-U9 Theorem 3.1. The selection $q_\Phi =+ 1$ (Theorem

S10.4) is the **Φ field's own U(1)_Y charge**, not the electron's electric charge, which is instead derived through ZS-U9 Theorem T3 via the neutral-Higgs fixing (inherited from ZS-U9 NC-U9.4).
NC-S10.9. ZS-S10 does NOT derive the SU(5) GUT unification scale or the GUT coupling. The Stückelberg mass scale $m_B \approx 7 \times 10^{16}$ GeV is computed using the low-energy value $g_Y \approx 0.345$ at M_Z; full RG running from M_Z to the Planck scale is outside the present scope (inherited from ZS-U9 NC-U9.3).
NC-S10.10. The 4π closure realization of Theorem S10.5-BPS is a statement about the **structure of the Z-mediated CPTP channel** in the ZS-S10 action, observable on Z-Spin quantum hardware via F-A7.3 (ZS-A7 TESTABLE). It is NOT an experimental claim about astrophysical BH horizons directly; the connection between hardware-testable ũ_seam periodicity and astrophysical BH observables remains OS-A7.2 OPEN (inherited from ZS-A7).

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§9. Falsification Gates

Five falsification gates are registered for ZS-S10:
F-S10.1 [MATH, DECISIVE]. If backward compatibility fails at any of the four verified regimes (B_μ = 0 recovery of ZS-F1, galactic screening of Stückelberg mass, inflation preservation of r = 0.0089, vortex core realization of Cor IV (F)), ZS-S10 is falsified. Current status: PASS (4/4, §5.2-5.6).
F-S10.2 [OBS, DECISIVE]. If the ZS-A7 F-A7.3 experimental gate returns 2π periodicity in the signed seam witness $\tilde{u}_{seam}$ (rather than the predicted 4π), ZS-S10 Theorem S10.5-BPS is falsified (as is ZS-A7 §3). Current status: OPEN, hardware-ready 2026-2028.
F-S10.3 [OBS, DECISIVE]. If any dark matter particle is detected at predicted mass/cross-section consistent with WIMP, axion, or sterile neutrino candidates [ZS-A5 §8 F-A5.7], the Corollary IV (B) Goldstone interpretation of the galactic halo is falsified, including ZS-S10 Theorem S10.3's IR limit. Current status: OPEN, inherited from ZS-A5.
F-S10.4 [MATH, DECISIVE]. If any of the five Standard Model anomaly cancellation conditions A1-A5 fails at integer precision in the ZS-S10 extended action, Theorem S10.2 Step 4 is falsified. Current status: PASS (5/5, ZS-U9 §7 preserved unchanged).
F-S10.5 [OBS, MODIFICATION REQUIRED]. If precision measurements of the photon mass exceed $m_\gamma < 10^{-18}$ eV (current PDG bound) in a way inconsistent with the Stückelberg mechanism (e.g., a fifth-force signature at laboratory scales), the ZS-S10 Stückelberg mass structure must be revised. Current status: OPEN but structurally consistent with all existing bounds.

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§10. Conclusion

ZS-S10 closes Gap G1 of the Trinity Braiding Theorem (ZS-U9) at the action level through a minimal extension of the ZS-F1 action: the partial derivative of the Z-bias field Φ is promoted to a gauge-covariant derivative $D_\mu \Phi = \left(\partial_\mu - i\kappa g_Y B_\mu\right)\Phi$, with mixing scale $\kappa = \sqrt{A/Q}$ uniquely determined by the Register-Total Normalization Theorem of ZS-M6. No new parameters, no new fields, no new postulates are introduced. The same minimal coupling simultaneously provides the action-level realization of ZS-A7 Corollary IV (Vortex Bose/Fermi Duality): the UV limit (vortex core) realizes the (F) branch with j = 1/2

spinor 4π closure, while the IR limit (galactic scale) realizes the (B) branch with 2π Goldstone winding. Both limits share the common Z-Anchor boundary condition $\left|\Phi\left(x_0\right)\right| = 0$.

Five theorems establish the mathematical content:

Theorem S10.1 (Stückelberg Mixing Scale, DERIVED-CONDITIONAL): $f^2 = \kappa^2 M_P^2 = (A/Q)M_P^2 = (35/4807)M_P^2$, unique at 10^{-14}% precision against alternative candidates.

Theorem S10.2 (L_XY = 0 Preservation, PROVEN-PERTURBATIVE): The sector independence of the ZS-F1 action is preserved to all orders in perturbation theory under the ZS-S10 extension, through a five-step proof combining Lorentz algebra decomposition, action-level absence, Ward-Takahashi identity, anomaly-free verification (including all five ZS-U9 SM anomaly conditions), and independent Schur protection.

Theorem S10.3 (Stückelberg-Cor IV Bridge, DERIVED): The UV (vortex core) and IR (galactic scale) limits of the ZS-S10 master action realize the (F) and (B) branches of ZS-A7 Corollary IV respectively, with explicit backward compatibility verified in four regimes (B_μ = 0 → ZS-F1; galactic → ZS-A1; inflation → ZS-U1; vortex core → Cor IV (F)).

Theorem S10.4 (Φ Field U(1)_Y Charge, DERIVED): $q_\Phi = +1$ uniquely fixed by four independent constraints (ZS-F1 §3.2 corpus-LOCKED convention; ZS-U9 Theorem 3.1 integer quantization; ZS-A6 n=1 vortex observational selection; EFT kinematic bound), with circularity resolved because Paths I-III are independent of ZS-S10 itself.

Theorem S10.5-BPS (Bogomolnyi BPS Spinor Lift, DERIVED): Seven-step explicit proof that the ZS-A7 Theorem 3.2-bis Kraus-operator 4π closure emerges naturally from the ZS-S10 action content at the vortex core, with observable realization $\tilde{u}_{seam}(\alpha) = cos(\alpha/2)$ matching F-A7.3 experimental prediction.

Gap G1 is therefore CLOSED at the action level. Gap G2 (ZS-M9 Table 2 assignment upgrade) is advanced through the anomaly-preserving structure of Theorem S10.2 but remains OPEN as a separate closure problem. The Trinity Braiding Theorem status is advanced from DERIVED (conditional on G1, G2) toward DERIVED (conditional on G2 only), pending confirmation of Theorem S10.4's Path I interpretation of ZS-F1 §3.2 as a corpus-LOCKED charge-1 convention.

The principal structural lesson is that **the integration of two previously independent corpus structures** — the compact U(1)_Z of ZS-F1 and the Standard Model hypercharge U(1)_Y — does not require new physics. A single minimal coupling $D_\mu \Phi$ with mixing scale $\kappa^2 = A/Q$ (already established by ZS-M6) is sufficient. The five theorems of this paper converge on a single statement: U(1)_Z and U(1)_Y are the same gauge group, and the ZS-S10 master action is the minimal action-level object that makes this identification manifest.

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Code Availability

The companion verification suite zs_s10_verify_v1_0.py is self-contained and publicly available at github.com/KennyKang-git/zspin/verify_scripts/. Execution: python3 zs_s10_verify_v1_0.py. Expected output: 36/36 PASS. Exit code: 0 (all pass) or 1 (any fail). Results are saved to zs_s10_v1_0_verification_results.json. Dependencies: numpy, scipy, mpmath (required). The suite uses mpmath (80-digit working precision, 50-digit display) for exact rational arithmetic and algebraic identities; numpy for matrix exponentials and Pauli algebra. No external data files required.

Appendix A. Verification Suite Summary

Target: 36/36 PASS across seven categories (A-G).
Category A: Foundational Constants (5 tests). Verify A = 35/437, Q = 11, (Z, X, Y) = (2, 3, 6), κ^2 = A/Q = 35/4807, M_P = 2.435 × 10^{18} GeV all inherited correctly from locked corpus.
Category B: Theorem S10.1 — Stückelberg Mixing Scale (6 tests). Verify $f^2 = \kappa^2 M^2_P$ exact rational identity; numerical uniqueness (A/Q vs A/(Q−Z), 3A/Q^2, A alone — all deviations reproduced at claimed precision); Stückelberg mass m_B for q_Φ = 1; lepton Stückelberg mass m_B^(L) at Y_L = −1/2; GUT-scale range (10^{16} GeV $\leq m_B \leq 10^{17}$ GeV); RG running consistency with ZS-S1 $\sin^2\theta$_W at integer arithmetic.
Category C: Theorem S10.2 — L_XY = 0 Preservation (7 tests). Verify each of the five proof steps: Lorentz algebra unchanged, action-level absence of direct X-Y terms, Ward-Takahashi identity preserved, anomaly-free verification (5/5 conditions PASS), Schur protection layer unchanged. Additional tests: Feynman vertex enumeration (new ZS-S10 vertices: B-B mass, θ-B mixing, fermion-B); κ^2 coefficient in θ-B mixing vertex.
Category D: Theorem S10.3 — Stückelberg-Cor IV Bridge (6 tests). Verify B_μ = 0 limit recovers ZS-F1 exactly (action-level equality); galactic Stückelberg screening at $e^{-10^{53}}$ suppression; inflation $|\Phi| \gg 1$ regime preserves ZS-U1 observables; vortex core Corollary IV (F) realization; galactic halo Corollary IV (B) realization; common Z-Anchor boundary condition.
Category E: Theorem S10.4 — q_Φ Charge (4 tests). Verify Path I (ZS-F1 §3.2 coefficient = 1); Path II (integer quantization modulo 2π); Path III ($n = 1$ vortex observational consistency); Path IV (EFT kinematic bound $1 \leq q_\Phi \leq 11$).
Category F: Theorem S10.5-BPS — BPS Spinor Lift (7 tests). Verify vortex existence (Jaffe-Taubes via ZS-A6 c_cigar = 0.06606); Frobenius expansion $\Phi \approx c_1(x + iy)$ near core; complex structure $J = -i\sigma_y$ (direct matrix identity); vector rep $R(\alpha) = exp(-i\alpha\sigma_y)$; spinor rep $U_Z(\alpha) = exp(-i\alpha\sigma_y/2)$; 4π closure $K_z(\alpha + 2\pi) = -K_z(\alpha)$; signed seam witness $\tilde{u}_{seam}(\alpha) = cos(\alpha/2)$.
Category G: Cross-Paper Consistency (1 test). Verify ZS-S10 imports match upstream values at integer arithmetic precision for: A (ZS-F2), Q (ZS-F5), κ^2 (ZS-M6), L_XY = 0 (ZS-F1, ZS-S1), j=1/2 uniqueness (ZS-M3), Kraus count = 2 (ZS-A7, ZS-A4), cigar vortex c_cigar (ZS-A6), Corollary IV (ZS-A7), anomaly conditions (ZS-U9), Higgs VEV/Weinberg angle/strong coupling (ZS-S4/S1).
Expected total: 36 tests, target 36/36 PASS.

Appendix B. Cross-Reference Table

Result	Source	Used In	Relation
A = 35/437	ZS-F2 v1.0 §7 (LOCKED)	S10.1	Input
Q = 11	ZS-F5 v1.0 §4 (PROVEN)	S10.1	Input
ZS-F1 action	ZS-F1 v1.0 §3.1 (PROVEN)	§3	Base action for extension
U(1)_Z gauge invariance	ZS-F1 v1.0 §3.2 (PROVEN)	S10.4 Path I	Corpus-LOCKED convention
$\kappa^2 = A/Q$	ZS-M6 Theorem 2.2.1 (DERIVED)	S10.1	Mixing scale basis
L_XY = 0	ZS-F1 §9, ZS-S1 §4 (PROVEN)	S10.2	Extended to ZS-S10
Continuum Perturbative Protection	ZS-M6 §7A (PROVEN-PERT)	S10.2 Steps 1-4	Extended to ZS-S10
Adjoint Obstruction Theorem	ZS-F2 §4.2A (PROVEN)	S10.2 Step 5	Schur protection layer
Z-Anchor $\lvert\Phi(x_0)\rvert = 0$	ZS-F1 §5.2 (PROVEN)	S10.3, S10.5-BPS	Common BC
Cigar vortex profile	ZS-A6 §4.5.6 (DERIVED)	S10.5-BPS Step 1	Existence/uniqueness
Frobenius expansion	ZS-A6 §4.4.1, §4.5.4 (DERIVED)	S10.5-BPS Step 2	Linear core behavior
ε-Halo profile	ZS-A1 §2.2 (DERIVED)	S10.3 IR limit	(B) branch realization
Goldstone winding	ZS-A6 §4.4.2 (PROVEN)	S10.3 IR limit	2π winding
Corollary IV	ZS-A7 §4.4 (DERIVED)	S10.3, S10.5-BPS	Target structure
Theorem 3.2-bis	ZS-A7 §3.2-bis (PROVEN)	S10.5-BPS Steps 5-6	Kraus 4π closure
j=1/2 uniqueness	ZS-M3 Theorem 5.1 (PROVEN)	S10.5-BPS Step 7	Uniqueness
SU(2) center	ZS-M3 Lemma 10.1 (PROVEN)	S10.5-BPS Step 6	$D^{1/2}(-I) = -I$
Half-angle V_XZ	ZS-F4 §7/§7B (DERIVED)	S10.5-BPS §5	Kraus square root origin
Stinespring/Kraus count	ZS-Q1 §3.3 (PROVEN)	S10.5-BPS Step 5	Kraus count = dim(Z) = 2
Trinity Braiding hypercharges	ZS-U9 Theorem 6.1 (DERIVED)	S10.2 Step 4, §6.5	Y_f inputs
Compact Phase Integer Quantization	ZS-U9 Theorem 3.1 (PROVEN)	S10.4 Path II	Integer $c \in \mathbb{Z}$
SM Anomalies A1-A5	ZS-U9 §7 (PROVEN)	S10.2 Step 4	Preserved unchanged

Result	Source	Used In	Relation
Higgs VEV	ZS-S4 §6.12 (DERIVED)	NC-S10.5	Preserved
$\sin^2\theta_W = 48/91 \cdot x^*$	ZS-S1 §8.2 (PROVEN)	§3.4	Input to g_Y estimate
$\alpha_s = 11/93$	ZS-S1 §8.1 (DERIVED)	Appendix A Category G	Cross-check
$\lambda_{vac} = 2A^2$	ZS-U5 §15.4 (DERIVED-COND)	§2, §7.2	Cigar vortex potential
$m_\rho = 2A \cdot M_P$	ZS-F1 §4.4 (DERIVED)	§5.4 inflation	Radial mode mass

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References

Internal (Z-Spin series)

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[Z2] K. Kang, "ZS-F2: Geometric Impedance A = 35/437," v1.0 (2026). [LOCKED A = 35/437; §4.2A Adjoint Obstruction Theorem PROVEN]

[Z3] K. Kang, "ZS-F4: Curvature Distribution and Holonomy," v1.0 (2026). [§7/§7B half-angle amplitudes V_{XZ}, V_{ZY} DERIVED]

[Z4] K. Kang, "ZS-F5: Gauge Symmetry Constraint — Why Q = 11," v1.0 (2026). [PROVEN (Z,X,Y)=(2,3,6); dim(Z)=2]

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[Z6] K. Kang, "ZS-M3: Regge-Holonomy, Immirzi & Z-Telomere," v1.0 (2026). [Theorem 5.1 j=1/2 PROVEN; Lemma 10.1 SU(2) sign flip PROVEN]

[Z7] K. Kang, "ZS-M6: Block-Laplacian Spectral Verification," v1.0 (2026). [Theorem 2.2.1 $\kappa^2=A/Q$ DERIVED; §7A Continuum Perturbative Protection Theorem PROVEN]

[Z8] K. Kang, "ZS-M9: McKay Correspondence and SM Field Classification," v1.0 (2026). [§5.2 McKay bridge DERIVED; Table 2 HYPOTHESIS strong]

[Z9] K. Kang, "ZS-S1: Gauge Coupling Unification," v1.0 (2026). [§4 $L_{XY} \equiv 0$ PROVEN; §8 α_s, $\sin^2\theta_W$]

[Z10] K. Kang, "ZS-S4: Electroweak & Higgs Completion," v1.0 (2026). [§6.12 Higgs VEV DERIVED at 0.12%]

[Z11] K. Kang, "ZS-U1: ε-Field Inflation," v1.0 (2026). [r = 0.0089 DERIVED]

[Z12] K. Kang, "ZS-U5: Quantum Gravity Bridge," v1.0 (2026). [§15.4 $\lambda_{vac} = 2A^2$ DERIVED-CONDITIONAL]

[Z13] K. Kang, "ZS-U9: Hypercharge Trinity," v1.0 (2026). [Theorem 6.1 DERIVED; Trinity Braiding; §7 anomalies 5/5 PASS]

[Z14] K. Kang, "ZS-A1: Galactic Dynamics & Morphology," v1.0 (2026). [§2 Goldstone halo DERIVED; 78/78 PASS]

[Z15] K. Kang, "ZS-A4: Black Hole Information and Quantum Protocol," v1.0 (2026). [Appendix A Z-sector basis $|4\rangle$, $|6\rangle$]

[Z16] K. Kang, "ZS-A6: Boundary Physics," v1.0 (2026, April 2026 update). [§4.5.6 cigar bounce DERIVED at 0.089% match]

[Z17] K. Kang, "ZS-A7: Horizon as Spinor — BH/WH Duality and the 4π Closure," v1.0.1 (2026). [§3 Horizon Spinor Theorem DERIVED; §3.2-bis Theorem 3.2-bis PROVEN; §4.4 Corollary IV DERIVED]
[Z18] K. Kang, "ZS-Q1: Geometric Decoherence and CPTP Channel," v1.0 (2026). [§3.3 Stinespring + Kraus PROVEN]

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Version History

v1.0 (April 2026): Initial public release. Consolidated from internal Z-Spin Collaboration research notes and four-session free-exploration development (April 2026) covering: Theorem S10.1 (Stückelberg mixing scale, DERIVED-CONDITIONAL via ZS-M6 Theorem 2.2.1), Theorem S10.2 ($L_{XY}=0$ preservation, PROVEN-PERTURBATIVE via five-step extension of ZS-M6 §7A), Theorem S10.3 (Stückelberg-Cor IV Bridge, DERIVED with backward compatibility verified in four regimes), Theorem S10.4 (Φ field $U(1)_Y$ charge $q_\Phi = +1$, DERIVED via four independent paths with circularity resolution), Theorem S10.5-BPS (Bogomolnyi BPS spinor lift, DERIVED via seven-step explicit proof with observable $\tilde{u}_{seam} = \cos(\alpha/2)$).
36 verification tests across seven categories (Foundational / S10.1 / S10.2 / S10.3 / S10.4 / S10.5-BPS / Cross-Paper). Five falsification gates (F-S10.1 through F-S10.5) registered. Ten non-claims (NC-S10.1 through NC-S10.10) registered to prevent overclaim.
Gap G1 of the Trinity Braiding Theorem (ZS-U9 NC-U9.2) CLOSED at the action level via ZS-S10 master action. Gap G2 (ZS-M9 Table 2 upgrade) advanced through anomaly-preserving structure but remains OPEN as a separate closure problem.
Zero new free parameters; A = 35/437 remains the sole geometric input. All upstream results preserved unchanged: ZS-F1 action ($B_\mu=0$ limit), ZS-A1 78/78 PASS (galactic screening), ZS-U1 r=0.0089 (inflation), ZS-A7 Corollary IV (F)/(B) duality (action-level realization).